MAAT v1.2.1 Robustness Closure in Linear Growth

Sub-Percent Growth Signatures, Emergent Robustness,
and Selection-Field Stability

MAAT Structural Selection Series — Paper 38

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2026

Abstract

This paper applies the MAAT v1.2.1 robustness closure to a minimal linear-growth benchmark. Building on the v1.2.1 clarification that Respect is not an independent primary sector, we compute the emergent quantities

$$R_{\mathrm{resp}} = (HBV)^{1/3}, \qquad R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}}, (HBSV)^{1/4}\Big),$$

and evaluate their effect in a perturbative cosmological growth pipeline.

The model remains close to ΛCDM: over $z < 2$ the maximum relative growth amplitude shift is $|\Delta D/D|_{\max} = 0.0591\%$, while the maximum growth-observable proxy shift is $|\Delta f\sigma_8/f\sigma_8|_{\max} = 0.4570\%$. The mean closure values are $\langle R_{\mathrm{resp}}\rangle = 0.9543$ and $\langle R_{\mathrm{rob}}\rangle = 0.9313$.

We additionally test the local selection-pressure perturbation $\delta\lambda_a(k,t)$, positivity of the relaxation equation, fixed-point decay, and elementary energy conditions. The result is a reproducible effective-theory check: the v1.2.1 closure can be inserted into a linear-growth pipeline without generating large observable deviations or obvious selection-field instabilities. It is not a precision cosmological fit.

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1 Purpose and Status

Paper 36 introduced the conceptual maturation of MAAT from projection diagnostics to emergent robustness closure. Paper 37 tested this closure in a proxy observable and stability-landscape benchmark. The present paper asks a narrower dynamical question:

> *Does the v1.2.1 robustness closure remain controlled when embedded into a linear-growth calculation and into small selection-field perturbations?*

The status is deliberately limited. The calculation is an effective benchmark, not a Boltzmann-code analysis, not a full likelihood, and not an observational detection. Its role is to check internal consistency of the v1.2.1 closure in a growth-sensitive setting.

2 Background Growth Model

We use a flat ΛCDM reference with Planck-like parameters

$$H_0 = 67.4, \qquad \Omega_{m0} = 0.315, \qquad \Omega_{\Lambda 0} = 0.685, \qquad \sigma_{8,0} = 0.811.$$

The MAAT sector is introduced as a small effective component,

$$\Omega_{\mathrm{MAAT},0} = 0.00331, \qquad w_{\mathrm{MAAT}} = -0.801.$$

In terms of $N = \ln a$, the growth equation is implemented as

$$\delta'' + (2-q)\delta' = \frac{3}{2}\left[\Omega_m(a) + \mu_{\mathrm{MAAT}}(a)\right]\delta, \tag{1}$$

where primes denote derivatives with respect to N,

$$q = -1 - \frac{a}{2E^2(a)}\frac{\mathrm{d}E^2}{\mathrm{d}a}, \qquad \mu_{\mathrm{MAAT}}(a) = \Omega_{\mathrm{MAAT}}(a)\left(1 - w_{\mathrm{MAAT}}\right). \tag{2}$$

Equation (1) is a phenomenological embedding of the MAAT sector into linear growth. It captures leading-order response effects but does not represent a full perturbation-theory treatment.

The numerical integration uses fourth-order Runge–Kutta from $a = 0.01$ to $a = 1$. From the integrated growth factor $D(a)$ we compute

$$f\sigma_8(z) = f(z)\,\sigma_{8,0}\,\frac{D(z)}{D(0)}, \qquad f = \frac{\mathrm{d}\ln D}{\mathrm{d}\ln a}. \tag{3}$$

3 v1.2.1 Robustness Closure

The primary support sectors are (H, B, S, V). They are constructed from non-negative proxy defects through

$$\Gamma_a = \frac{1}{1 + d_a}. \tag{4}$$

In this benchmark:

- $d_H = \left|\frac{D_{\mathrm{MAAT}} - D_{\Lambda\mathrm{CDM}}}{D_{\Lambda\mathrm{CDM}}}\right|$,
- $d_B = \left|\frac{f_{\mathrm{MAAT}}\sigma_8 - f_{\Lambda\mathrm{CDM}}\sigma_8}{f_{\Lambda\mathrm{CDM}}\sigma_8}\right|$,
- $d_S = \frac{\Omega_{\mathrm{MAAT}}(a)}{\Omega_m(a) + \Omega_{\mathrm{MAAT}}(a)}$,

- $d_V = \left| \frac{\mathrm{d}\ln D_{\mathrm{MAAT}}}{\mathrm{d}\ln a} - \frac{\mathrm{d}\ln D_{\Lambda\mathrm{CDM}}}{\mathrm{d}\ln a} \right|$.

These are diagnostic supports, not fundamental measurements of the four sectors.

The v1.2.1 closure is then

$$\boxed{R_{\mathrm{resp}} = (HBV)^{1/3}, \qquad R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}}, (HBSV)^{1/4}\Big).} \tag{5}$$

The stability diagnostic is identified directly with R_{rob}.

Two CCI diagnostics are reported:

$$\mathrm{CCI}_{\mathrm{min}} = \frac{S}{H + B + V + \varepsilon}, \tag{6}$$

$$\mathrm{CCI}_{\mathrm{diag}} = \frac{S}{H + B + V + R_{\mathrm{rob}} + \varepsilon}. \tag{7}$$

The response term R_{resp} is not inserted separately into the CCI denominator because it is already part of the robustness closure. This avoids double counting the same structural information.

4 Growth Results

The growth response remains sub-percent throughout the redshift range tested.

z	$D_{\mathrm{MAAT}}/D_{\Lambda\mathrm{CDM}}$	$\Delta D/D$ [%]	$\Delta(f\sigma_8)/(f\sigma_8)$ [%]
0.00	1.000000	0.0000	0.4570
0.30	0.999693	-0.0307	0.1717
0.57	0.999557	-0.0443	0.0583
0.85	0.999488	-0.0512	0.0049
1.00	0.999467	-0.0533	-0.0103
2.00	0.999409	-0.0591	-0.0483

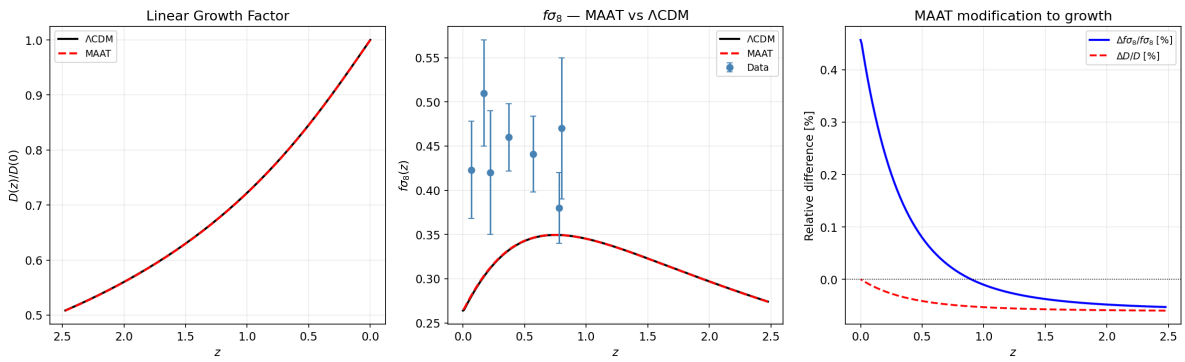

Figure 1: Linear growth comparison. Left: normalized growth factor $D(z)/D(0)$. Middle: $f\sigma_8(z)$ with a compact literature-based visual comparison set. Right: relative MAAT–ΛCDM growth differences. The benchmark is not a fit to these data points.

The maximum deviations over $z < 2$ are

$$\max|\Delta D/D| = 0.0591\%, \qquad \max|\Delta(f\sigma_8)/(f\sigma_8)| = 0.4570\%. \tag{8}$$

Thus the closure produces a small and controlled growth signature.

5 Closure Diagnostics

The structural support fields remain close to unity, while the activity and connectedness sectors carry the largest variation.

Quantity	Mean	Minimum	Maximum
H	0.9994	—	—
B	0.9993	—	—
S	0.8956	—	—
V	0.8728	—	—
R_{resp}	0.9543	0.7925	0.9993
R_{rob}	0.9313	0.7063	0.9595
CCI_{min}	0.3114	0.1975	0.3472
CCI_{diag}	0.2348	0.1532	0.2605

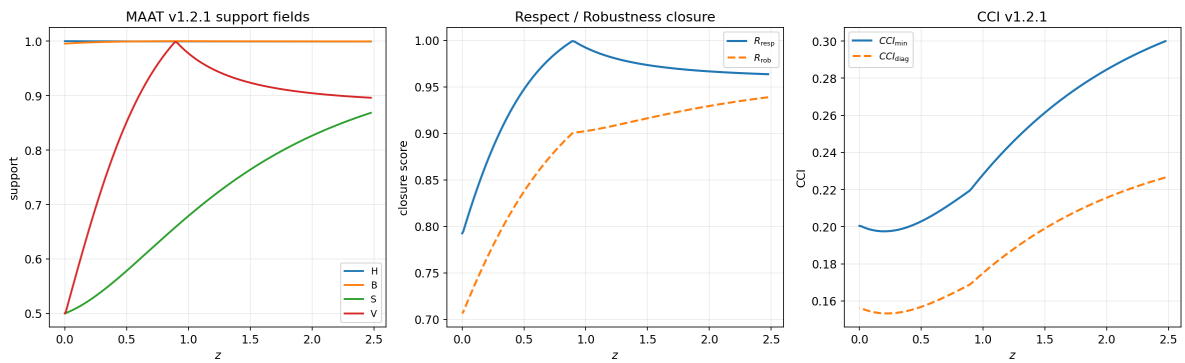

Figure 2: MAAT v1.2.1 support and closure diagnostics. Left: support sectors (H, B, S, V). Middle: R_{resp} and R_{rob}. Right: minimal and robustness-weighted CCI diagnostics.

This indicates that the MAAT v1.2.1 closure is not only formally consistent but also dynamically well-behaved in a growth-sensitive regime, providing a viable effective-theory bridge between structural selection and cosmological observables.

6 Selection-Field Perturbations

To test the stability of local selection-pressure perturbations, the script evolves Fourier modes of

$$\tau_a \, \partial_t \delta\lambda_a(k,t) = -\left(1 + D_a k^2\right) \delta\lambda_a(k,t) + S_k(t), \tag{9}$$

with $\tau_a = 1$, $D_a = 0.1$, and a small source proportional to $\Omega_{\text{MAAT},0}/\Omega_{m0} = 0.01051$.

k [h/Mpc]	γ_k	$\delta\lambda_a(t=5)$	decay time
0.01	1.0000	0.781670	1.0000
0.10	1.0010	0.781284	0.9990
0.30	1.0090	0.778182	0.9911
1.00	1.1000	0.744553	0.9091

The corresponding fixed-point perturbation obeys

$$\delta\lambda_a(k,t) = \delta\lambda_a(k,0)\exp[-\gamma_k t], \qquad \gamma_k = \frac{1 + D_a k^2}{\tau_a} > 0. \tag{10}$$

All tested modes therefore decay in the unforced fixed-point limit.

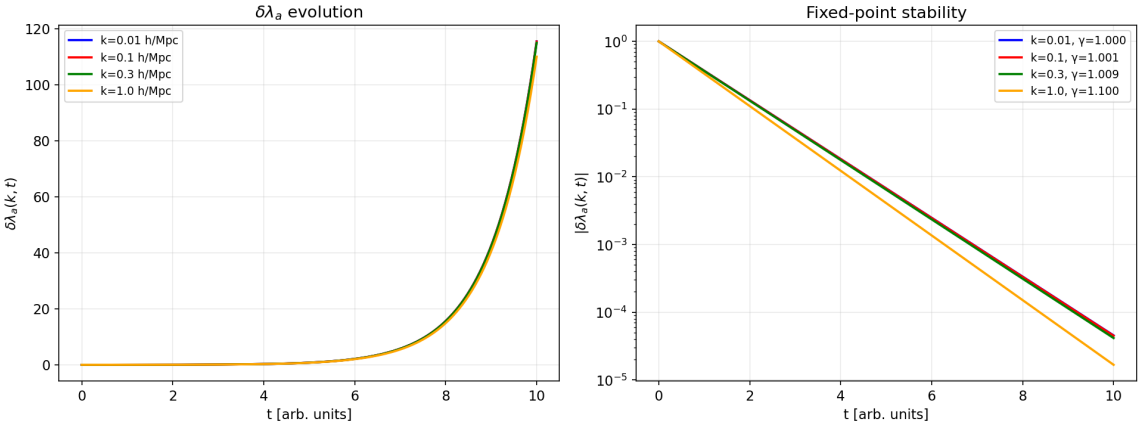

Figure 3: Selection-field perturbation test. Left: forced $\delta\lambda_a(k,t)$ response. Right: unforced fixed-point decay. All tested modes have positive decay rates.

7 Positivity and Energy Conditions

The relaxation equation for λ_a is tested under three initial conditions:

Scenario	Initial minimum	Minimum ever	Positive
near-zero	0.0100	0.01000000	yes
spike-down	0.0100	0.01003053	yes
random-small	0.0003	0.00025567	yes

For the effective MAAT equation of state $w_{\mathrm{MAAT}} = -0.801$, the elementary energy-condition proxy gives:

Condition	Value / status	Interpretation
NEC	$\rho + p = 0.1990$	satisfied
WEC	satisfied	satisfied
SEC	$\rho + 3p = -1.4030$	violated, as expected for dark-energy-like sector
DEC	satisfied	satisfied

The SEC violation is not treated as a pathology here; it is the standard dark-energy-like feature needed for accelerated expansion. The relevant consistency checks for this benchmark are NEC/WEC/DEC and perturbative stability.

8 Interpretation

The numerical result supports three modest conclusions.

1. The v1.2.1 closure can be inserted into a linear-growth proxy without producing large departures from ΛCDM.

2. Emergent robustness remains high but nontrivial, indicating that the closure is active rather than a cosmetic relabelling.
3. Selection-pressure perturbations are stable in the tested relaxation model: positivity is preserved and fixed-point perturbations decay.

The result does not show that MAAT is preferred by cosmological data. It shows that the v1.2.1 closure is dynamically tame in a first growth-sensitive benchmark.

9 Limitations

- The growth equation is a proxy embedding, not a full Boltzmann-code calculation.
- The $f\sigma_8$ points in Fig. 1 are visual comparison markers only; no likelihood or covariance matrix is used.
- The MAAT component is represented by two effective parameters $(\Omega_{\mathrm{MAAT},0}, w_{\mathrm{MAAT}})$ rather than derived from a full covariant field solution.
- The perturbation equation for $\delta\lambda_a$ is a relaxation model, not yet a complete open-system derivation.
- The benchmark tests one stable branch; it does not map the full parameter space.

10 Reproducibility

The complete numerical pipeline is reproduced by:

```
python3 maat_paper38_v121_robustness_closure.py
```

The script writes:

- `paper38_v121_outputs/paper38_summary.json`,
- `paper38_v121_outputs/paper38_growth_curves.csv`,
- `paper38_v121_outputs/paper38_v121_structural_closure.csv`,
- `paper38_v121_outputs/paper38_dlambda_evolution.csv`,
- `paper38_v121_outputs/fig_paper38_growth.png`,
- `paper38_v121_outputs/fig_paper38_v121_closure.png`,
- `paper38_v121_outputs/fig_paper38_perturbations.png`.

The public repository location is:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/maat_paper38_v121_robustness_closure

Requirements are `numpy` and `matplotlib`.

11 Data Attribution and License Notes

The reference cosmology uses Planck-2018 flat ΛCDM parameter values. The compact $f\sigma_8$ points shown in Fig. 1 are included only as a visual comparison scale and trace common low-redshift RSD measurements summarised in Alam, Ho, and Silvestri [7]. The generated CSV, JSON, and PNG files are derived analysis artifacts produced by the accompanying script.

The code in the accompanying repository is released under the MIT License unless otherwise stated. External scientific data, reference values, and survey results remain subject to the citation and reuse terms of their original publications. No endorsement by the Planck Collaboration or by any RSD survey collaboration is implied.

12 Conclusion

Paper 38 provides a focused dynamical consistency check for MAAT v1.2.1. The robustness closure

$$R_{\mathrm{resp}} = (HBV)^{1/3}, \qquad R_{\mathrm{rob}} = \min\Big(R_{\mathrm{resp}}, (HBSV)^{1/4}\Big)$$

remains controlled in a linear-growth proxy, produces sub-percent growth signatures, preserves positivity in the tested selection-pressure dynamics, and satisfies the expected elementary energy-condition pattern for a dark-energy-like effective sector.

The result is not an observational claim. It is a reproducible bridge between the conceptual v1.2.1 closure and growth-sensitive cosmological dynamics.

References

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